

Section 27: Port Security

Layer: L2 | Reference: Odom Vol2 Ch11-12: Port Security, DHCP Snooping | *Original study notes — CCNA 200-301*

Port Security restricts which MAC addresses are allowed to send traffic on a switch port, protecting against MAC flooding attacks and unauthorized device connections. It is configured per-port and is a staple CCNA security topic.

Key Concepts

- Port security only works on access ports (or ports manually set to access mode) - not on dynamic/trunk ports by default
- Maximum MAC addresses per port defaults to 1 if port security is enabled without specifying a max
- Secure MAC learning modes: static (manually configured), dynamic (learned, lost on reboot), sticky (learned, but saved to running-config and persists across reboots if saved)
- Violation modes: protect (drop offending traffic silently, no log), restrict (drop traffic, log it, increment violation counter), shutdown (default - err-disable the entire port)
- An err-disabled port from a security violation must be manually recovered with shutdown / no shutdown unless errdisable recovery is configured
- errdisable recovery cause psecure-violation enables automatic recovery after a configurable interval
- Port security pairs naturally with DHCP snooping: DHCP snooping builds a binding table (MAC-to-IP-to-port) that other features like DAI and IP Source Guard rely on
- DHCP snooping trusted ports (uplinks to the real DHCP server) are allowed to relay DHCP server messages (OFFER/ACK); untrusted ports (access ports) have those messages blocked
- DAI (Dynamic ARP Inspection) validates ARP packets against the DHCP snooping binding table to prevent ARP spoofing - requires DHCP snooping to be enabled first

Command Reference

Command	Purpose
<code>switchport port-security</code>	Enable port security on the interface (must be access mode first)
<code>switchport port-security maximum <n></code>	Set max number of allowed MAC addresses (default 1)
<code>switchport port-security mac-address sticky</code>	Learn MACs dynamically but save them into the config
<code>switchport port-security mac-address <mac></code>	Manually configure a static allowed MAC address
<code>switchport port-security violation {protect restrict shutdown}</code>	Set the action taken on a violation
<code>show port-security interface <intf></code>	View port security status, violation count, and learned MACs

Command	Purpose
<code>ip dhcp snooping</code>	Globally enable DHCP snooping
<code>ip dhcp snooping vlan <id></code>	Enable DHCP snooping for a specific VLAN
<code>ip dhcp snooping trust</code>	Mark an interface as trusted (interface config mode) - use on uplinks toward the real DHCP server
<code>ip arp inspection vlan <id></code>	Enable Dynamic ARP Inspection on a VLAN

Configuration Walkthrough

Basic port security: max 2 MACs, sticky learning, shutdown on violation

```
SW1(config)#interface FastEthernet0/5
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security maximum 2
SW1(config-if)#switchport port-security mac-address sticky
SW1(config-if)#switchport port-security violation shutdown
```

DHCP snooping with a trusted uplink

```
SW1(config)#ip dhcp snooping
SW1(config)#ip dhcp snooping vlan 10,20
SW1(config)#interface GigabitEthernet0/1
SW1(config-if)#ip dhcp snooping trust
```

Auto-recovery from err-disabled port security violations

```
SW1(config)#errdisable recovery cause psecure-violation
SW1(config)#errdisable recovery interval 300
```

Worked Scenario

Q: A port configured with 'switchport port-security violation restrict' keeps logging violations but traffic from the unauthorized device is still being dropped - is that expected?

Yes. 'Restrict' drops the offending traffic and logs/increments the violation counter, but does NOT take the port down, unlike 'shutdown' mode. This is the intended difference: restrict keeps the port operational for legitimate traffic while silently dropping violating frames; only shutdown mode err-disables the whole port.

Common Mistakes / Gotchas

- Trying to enable port security on a trunk port without first reconsidering whether that's really the design intent
- Forgetting that 'sticky' MACs still need 'copy run start' to survive a reload
- Setting violation mode to shutdown on a port expected to have frequent legitimate device swaps, causing unnecessary lockouts
- Enabling DAI without first enabling DHCP snooping - DAI depends on the snooping binding table

Exam Tips

- ★ *Know all three violation modes cold: protect (silent drop), restrict (drop+log), shutdown (err-disable, default)*
- ★ *Sticky = dynamic learning + persistent in config once saved*
- ★ *DHCP snooping trust goes on the uplink toward the legitimate DHCP server, not on access ports*